

DETAILS OF LIBRARY

LIBRARIES USED:

- NumPy
- SciPy
- Matplotlib
- scikit-learn
- TensorFlow
- TextBlob

- **NUMPY**

Why it is used:

Used for numerical computations, matrix operations, and handling multidimensional arrays efficiently.

Applications:

Data analysis, scientific computing, machine learning, linear algebra, and signal processing.

- **SCIPY**

Why it is used:

Used for advanced statistical functions and scientific calculations.

Applications:

Optimization, statistics, scientific simulations, image processing, and engineering computations.

- **MATPLOTLIB**

Why it is used:

Used for creating graphs and visualizing data in different chart formats.

Applications:

Data visualization, trend analysis, plotting histograms, scatter plots, line graphs, and dashboards.

- **SCIKIT-LEARN**

Why it is used:

Used for implementing machine learning algorithms and preprocessing techniques.

Applications:

Regression, classification, clustering, model evaluation, and predictive analytics.

- **TENSORFLOW**

Why it is used:

Used for deep learning and neural network-based machine learning tasks.

Applications:

Artificial intelligence, NLP, image recognition, deep learning, and recommendation systems.

- **TEXTBLOB**

Why it is used:

Used for natural language processing and sentiment analysis tasks.

Applications:

Sentiment analysis, text classification, spell correction, and language processing.